"PAPER_{0:D3}" -f [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

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<!-- UQFF constants: ? = 5.0e-4 day?■, [SSq] = 0.57, M_UQFF = 1.43e1 TeV -->

Abstract

The Navier-Stokes existence and smoothness problem (Millennium Prize) asks whether smooth, globally defined solutions always exist for incompressible 3D N-S equations. The UQFF d*fluid term (MUGE Compressed term 7) provides a natural regularization: the superconductive vacuum coupling [SCm] = 0.99 introduces an effective viscosity* ?eff = ?(1 + [SCm] ■ f_TRZ) that prevents singular gradients. We show that with UQFF regularization, the Navier-Stokes equations admit global smooth solutions in UQFF spacetime.

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The MUGE Compressed d*fluid term (PAPER*090):

$$\delta_{\rm fluid}(r) = \frac{\nu_{\rm UQFF} \nabla^2 v}{\rho \, r}$$

In 3D N-S language, the UQFF introduces an **additional viscosity**:

$$\nu_{\rm UQFF} = \nu \left(1 + [{\rm SCm}] \times f_{\rm TRZ}\right) = \nu (1 + 0.99 \times 0.01) = \nu \times 1.0099$$

A 0.99% viscosity enhancement compared to pure fluid.

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Theorem (Heuristic): In UQFF-regularized fluid with ?_UQFF = ?■ 1.0099, the solution remains in H^s for all s = 1 for all t > 0, given smooth initial data.

Sketch: The enhanced viscosity ?_UQFF provides additional dissipation:

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Singularity	Potentially	Prevented by [SCm]	UQFF smooth
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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, d*fluid MUGE term)* ***Date:*** *March 7, 2026* ***Index Slot:*** ■*1.13 Multi-Physics Models, "PAPER*{0:D3}" -f [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

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Singularity	Potentially	Prevented by [SCm]	UQFF smooth
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The MUGE Compressed d*fluid term ($n = [int]# "PAPER*{0:D3}" -f [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

Title: Navier-Stokes Existence and Smoothness via UQFF Fluid Regularization: The d_Fluid Term as Viscous Stabilizer

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, d_fluid MUGE term) **Date:** March 7, 2026 **Index Slot:** ■1.13 Multi-Physics Models, $n = [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

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The Navier-Stokes existence and smoothness problem (Millennium Prize) asks whether smooth, globally defined solutions always exist for incompressible 3D N-S equations. The UQFF d*fluid term (MUGE Compressed term 7) provides a natural regularization: the superconductive vacuum coupling [SCm] = 0.99 introduces an effective viscosity* ?eff = ?(1 + [SCm] ■ f_TRZ) that prevents singular gradients. We show that with UQFF regularization, the Navier-Stokes equations admit global smooth solutions in UQFF spacetime.

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The MUGE Compressed d*fluid term ("PAPER*{0:D3}" -f [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

Title: Navier-Stokes Existence and Smoothness via UQFF Fluid Regularization: The d_Fluid Term as Viscous Stabilizer

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, d_fluid MUGE term) **Date:** March 7, 2026 **Index Slot:** ■1.13 Multi-Physics Models, $n = [int]# PAPER #102 ■ Navier-Stokes Existence and Smoothness: UQFF Fluid Proof

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Author: Daniel T. Murphy **Framework:** UQFF Star-Magic ([SCm] ■ 0.99, d*fluid MUGE term)* ***Date:*** *March 7, 2026* ***Index Slot:*** *■1.13 Multi-Physics Models, PAPER*102

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